

# Justin Rhee

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## EDUCATION

### University of Colorado Boulder

*Bachelor of Science in Computer Science, Minor in Business | GPA: 3.888*

Boulder, CO

August 2024 – Present

## EXPERIENCE

### Software Engineering Intern

June 2025 – August 2025; June 2026 – Present

*Ecliptic Enterprises*

*Pasadena, CA*

- Developed and scaled a Raspberry Pi-based coelostat prototype into a networked dual-axis laser-tracking system
- Presented engineering projects to managers and clients, communicating performance and operational impact
- Extended summer internship into a remote, part-time software engineering role during the academic year

### Course Assistant - Data Science with Probability and Statistics

August 2025 – Present

*University of Colorado Boulder*

*Boulder, CO*

- Led office hours and review sessions to support a class of 160+ students in data science, probability, and statistics
- Clarified statistical concepts, debugging, and programming practices by answering technical questions on Piazza

### Business/IT Intern

May 2024 – August 2024

*Kids in Motion Pediatric Therapy*

*Torrance, CA*

- Automated data migration between the old and new electronic medical record (EMR) systems using Python in collaboration with the Clinic Director, saving 200+ hours of staff time

## PROJECTS

### Startracker Simulation | Python, PySide6, Serial I/O, Multithreading

August 2026

- Developed a Python-based hardware simulation and test automation tool to support TCB/ATB engineering validation while physical hardware was unavailable
- Built a PySide6 GUI for serial communication testing with configurable COM ports, baud rates, operational/testing modes, checksum controls, and live RX/TX terminal output
- Implemented packet generation, framed serial transmission, telemetry response handling, subscription requests, encoded packet testing, and invalid checksum/error-case simulation
- Integrated multithreading, serial I/O, checksum validation, and telemetry data management to create a repeatable pre-hardware test harness for engineering teams

### Dual-Axis Laser-Tracking Coelostat | Python, PySide6, OpenCV, Sockets

July 2025

- Designed a dual-axis motor control system for 360° laser tracking with real-time feedback and socket communication between software, cameras, and controllers, averaging a 9 Hz command cycle synced to step completion
- Built PySide6 GUI with dual-camera feeds, centroid overlay and status indicator, and real-time data visualization
- Achieved sub-degree tracking stability with  $\pm 0.05^\circ$  precision and  $\pm 0.1^\circ$  tolerance for consistently smooth motion
- Implemented PID control and low-pass filtering to stabilize motor response and enhance tracking accuracy
- Applied HSV segmentation, contour analysis, and morphological filtering to track centroids under variable lighting
- Used multithreading for dual-camera coordination, processing, motion, and data tasks, reducing feedback latency

### Document Automation System | Python, Tkinter, File Processing, Automation

August 2025

- Engineered a backend processor for placeholder replacement and structured logging across 100+ Office templates using control-file mapping, recursive format traversal, and version-safe output, cutting manual entry by  $\approx 70\%$
- Built a multi-stage Tkinter GUI for folder navigation, placeholder entry, and batch processing of Office templates
- Implemented importlib-based auto-installer for dependency setup across new systems
- Designed collapsible folder trees and summary views showing selected files and results for clearer navigation

## RELEVANT COURSEWORK

Data Structures   Discrete Structures   Computer Systems   Data Science and Statistics   Linear Algebra  
Algorithms   Intro to AI   Principles of Programming Languages   Database Systems   Generative AI  
Applied Machine Learning   Software Dev   Intro to Cybersecurity   Business and Financial Analytics

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript, SQL, Scala, HTML/CSS

**Technologies:** PySide6, OpenCV, pandas, NumPy, Git, GitHub, Docker, MySQL, Jupyter